

Euler 函数

定义 (Euler 函数) 对 $\forall n \in \mathbb{Z}_{\geq 1}$, 定义 $\varphi(n)$ 为不超过 n 而与 n 互素的正整数个数.

例: 对于 $1 \in \mathbb{Z}_{\geq 1}$, 不超过 1 而与 1 互素的正整数只有 1. $\therefore \varphi(1) = 1$

对于 $2 \in \mathbb{Z}_{\geq 1}$, 不超过 2 而与 2 互素的正整数只有 1. $\therefore \varphi(2) = 1$

对于 $3 \in \mathbb{Z}_{\geq 1}$, 不超过 3 而与 3 互素的正整数有: 1, 2. $\therefore \varphi(3) = 2$

对于 $4 \in \mathbb{Z}_{\geq 1}$, 不超过 4 而与 4 互素的正整数有: 1, 3. $\therefore \varphi(4) = 2$

对于 $5 \in \mathbb{Z}_{\geq 1}$, 不超过 5 而与 5 互素的正整数有: 1, 2, 3, 4. $\therefore \varphi(5) = 4$

对于 $6 \in \mathbb{Z}_{\geq 1}$, 不超过 6 而与 6 互素的正整数有: 1, 5. $\therefore \varphi(6) = 2$

对于 $7 \in \mathbb{Z}_{\geq 1}$, 不超过 7 而与 7 互素的正整数有: 1, 2, 3, 4, 5, 6. $\therefore \varphi(7) = 6$

对于 $8 \in \mathbb{Z}_{\geq 1}$, 不超过 8 而与 8 互素的正整数有: 1, 3, 5, 7. $\therefore \varphi(8) = 4$

对于 $9 \in \mathbb{Z}_{\geq 1}$, 不超过 9 而与 9 互素的正整数有: 1, 2, 4, 5, 7, 8. $\therefore \varphi(9) = 6$

对于 $10 \in \mathbb{Z}_{\geq 1}$, 不超过 10 而与 10 互素的正整数有: 1, 3, 7, 9. $\therefore \varphi(10) = 4$

对于 $11 \in \mathbb{Z}_{\geq 1}$, 不超过 11 而与 11 互素的正整数有: 1, 2, 3, 4, 5, 6, 7, 8, 9, 10. $\therefore \varphi(11) = 10$

对于 $12 \in \mathbb{Z}_{\geq 1}$, 不超过 12 而与 12 互素的正整数有: 1, 5, 7, 11. $\therefore \varphi(12) = 4$

对于 $13 \in \mathbb{Z}_{\geq 1}$, 不超过 13 而与 13 互素的正整数有: 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12.

$\therefore \varphi(13) = 12$

对于 $14 \in \mathbb{Z}_{\geq 1}$, 不超过 14 而与 14 互素的正整数有: 1, 3, 5, 9, 11, 13. $\therefore \varphi(14) = 6$

对于 $15 \in \mathbb{Z}_{\geq 1}$, 不超过 15 而与 15 互素的正整数有: 1, 2, 4, 7, 8, 11, 13, 14.

$\therefore \varphi(15) = 8$

Lemma: p 是任意的素数, 则有: $\varphi(p) = p-1$

Proof: $\because p$ 是任意的素数 $\therefore p \in \mathbb{Z}$ 且 $p \geq 2$. $\therefore p \in \mathbb{Z}_{\geq 1} \therefore \varphi(p)$ 有定义.

$\therefore$ 对 $\forall i=1, \dots, p-1$, 有: $\gcd(p, i) = 1$

$\therefore 1$ 与 p 互素, 2 与 p 互素, $\dots$, $p-1$ 与 p 互素.

$\therefore \gcd(p, p) = p \geq 2 \therefore p$ 与 p 不互素

$\therefore \varphi(p) = p-1$ $\square$

~~Lemma: $\lim_{n \rightarrow +\infty} \varphi(n) = +\infty$~~

~~Proof: 对 $\forall A > 0$, $\exists N \in \mathbb{Z}_{\geq 1}$, s.t. $N > A + 100$~~

Lemma: p 是任意的素数, 对 $\forall k \in \mathbb{Z}_{\geq 1}$, 有: $\varphi(p^k) = p^{k-1}(p-1)$

Proof: $\because p$ 是任意的素数 $\therefore p \in \mathbb{Z}$ 且 $p \geq 2$

$\therefore k \in \mathbb{Z}_{\geq 1} \therefore p^k \in \mathbb{Z}$ 且 $p^k \geq p^1 = p \geq 2$

$\therefore p^k \in \mathbb{Z}_{\geq 1}$, 且 p^k 就是 p^k 的素因子分解形式. d 是一个任意的非零整数, 则有:

$d \mid p^k \Leftrightarrow d = \pm p^\varepsilon$, 其中 $\varepsilon \in \mathbb{Z}_{\geq 0}$, $\varepsilon \leq k$

$\therefore p^k$ 的全部因数是: $\pm 1, \pm p, \pm p^2, \pm p^3, \dots, \pm p^{k-1}, \pm p^k$

~~对 $\forall \alpha \in \mathbb{Z}_{\geq 1}$~~ 对 $\forall \alpha \in \mathbb{Z}_{\geq 1}$, 有:

若 $p \mid \alpha$, 则有: $\because p \mid \alpha$ 且 $p \mid p^k \therefore \gcd(\alpha, p^k) \geq p \geq 2 > 1 \therefore \gcd(\alpha, p^k) \neq 1$

若 $\gcd(\alpha, p^k) \neq 1$, 则有: $\because \alpha \in \mathbb{Z}_{\geq 1}$ 且 $p^k \in \mathbb{Z}_{\geq 2} \therefore \gcd(\alpha, p^k) \in \mathbb{Z}_{\geq 1}$

$$\because \gcd(\alpha, p^k) \mid p^k \text{ 且 } \gcd(\alpha, p^k) \in \mathbb{Z}_{\geq 1}$$

$$\therefore \gcd(\alpha, p^k) \in \{1, p, p^2, p^3, \dots, p^{k-1}, p^k\}$$

$$\because \gcd(\alpha, p^k) \neq 1 \quad \therefore \gcd(\alpha, p^k) \in \{p, p^2, \dots, p^k\}$$

$$\therefore p \mid \gcd(\alpha, p^k) \quad \therefore \gcd(\alpha, p^k) \mid \alpha \quad \therefore p \mid \alpha$$

$$\therefore \text{对 } \forall \alpha \in \mathbb{Z}_{\geq 1}, \text{ 有: } p \mid \alpha \Leftrightarrow \gcd(\alpha, p^k) \neq 1 \Leftrightarrow \alpha \text{ 与 } p^k \text{ 不互素}$$

$$\therefore \text{不超过 } p^k \text{ 的正整数有 } 1, 2, \dots, p^k \text{ 这 } p^k \text{ 个.}$$

$$\text{在这 } p^k \text{ 个正整数中, 与 } p^k \text{ 不互素的正整数有 } p \times 1 = p, p \times 2 = 2p, p \times 3 = 3p, \dots, p \times p^{k-1} = p^k \text{ 这 } p^{k-1} \text{ 个}$$

$$\therefore \text{不超过 } p^k \text{ 而与 } p^k \text{ 互素的正整数有 } p^k - p^{k-1} \text{ 个.}$$

$$\therefore \varphi(p^k) = p^k - p^{k-1} \quad (\text{当 } k=1 \text{ 时, 就是 } \varphi(p) = p-1)$$

$$\therefore \varphi(p^k) = p^k - p^{k-1} = p^{k-1}(p-1) \quad \square$$

Lemma: n 是一个任意的正整数, 则 n 有素因子分解: $n = p_1^{a_1} \dots p_r^{a_r}$, 其中 $r \in \mathbb{Z}_{\geq 0}$

($r=0$ 对应于 $n=1$ 的情形), $p_1, \dots, p_r$ 是相异素数, $a_1, \dots, a_r \in \mathbb{Z}_{\geq 1}$.

$$\text{对 } \forall \alpha \in \mathbb{Z}_{\geq 1}, \text{ 有: } \gcd(\alpha, n) \neq 1 \Leftrightarrow p_1 \mid \alpha \text{ 或 } p_2 \mid \alpha \text{ 或 } \dots \text{ 或 } p_r \mid \alpha.$$

$$\text{proof: 当 } n=1 \text{ 时, } r=0. \text{ 对 } \forall \alpha \in \mathbb{Z}_{\geq 1}, \text{ 有: } \gcd(\alpha, n) = \gcd(\alpha, 1) = 1$$

下设 $n \in \mathbb{Z}_{\geq 2}$. 此时 $r \in \mathbb{Z}_{\geq 1}$.

$$\text{对 } \forall \alpha \in \mathbb{Z}_{\geq 1}. \text{ 若 } p_1 \mid \alpha \text{ 或 } p_2 \mid \alpha \text{ 或 } \dots \text{ 或 } p_r \mid \alpha \quad \therefore \exists p_\mu (\mu \in \{1, 2, \dots, r\}), \text{ 使得 } p_\mu \mid \alpha$$

$$\therefore p_\mu \mid n \text{ 且 } p_\mu \mid \alpha \quad \therefore \gcd(\alpha, n) \geq p_\mu \geq 2 > 1 \quad \therefore \gcd(\alpha, n) \neq 1.$$

若 $\gcd(\alpha, n) \neq 1$, 则有:

$$\because n \in \mathbb{Z}_{\geq 1} \quad \therefore \gcd(\alpha, n) \in \mathbb{Z}_{\geq 1} \quad \therefore \gcd(\alpha, n) \mid n$$

$$\therefore \gcd(\alpha, n) = p_1^{e_1} \cdots p_r^{e_r}, \text{ 其中 } e_1, \dots, e_r \in \mathbb{Z}_{\geq 0}, e_1 \leq a_1, \dots, e_r \leq a_r.$$

$$\therefore \gcd(\alpha, n) \neq 1 \quad \therefore e_1, \dots, e_r \text{ 不全为 } 0$$

(假设 $e_1, \dots, e_r$ 全为 0, 则有: $\gcd(\alpha, n) = p_1^{e_1} \cdots p_r^{e_r} = p_1^0 \cdots p_r^0 = 1 \times \cdots \times 1 = 1$ 矛盾.)

$$\therefore \exists \delta \in \{1, \dots, r\}, \text{ s.t. } e_\delta \neq 0 \quad \therefore e_\delta \in \mathbb{Z}_{\geq 1} \quad \therefore p_\delta \mid \gcd(\alpha, n)$$

$$\therefore p_\delta \mid \gcd(\alpha, n) \text{ 且 } \gcd(\alpha, n) \mid \alpha \quad \therefore p_\delta \mid \alpha$$

$$\therefore p_1 \mid \alpha \text{ 或 } p_2 \mid \alpha \text{ 或 } \cdots \text{ 或 } p_r \mid \alpha$$

$$\therefore \text{对 } \forall \alpha \in \mathbb{Z}_{\geq 1}, \text{ 有: } \gcd(\alpha, n) \neq 1 \iff p_1 \mid \alpha \text{ 或 } p_2 \mid \alpha \text{ 或 } \cdots \text{ 或 } p_r \mid \alpha \quad \square$$

Lemma ($\varphi(n)$ 的计算式) n 是一个任意的正整数, 则 n 有素因子分解: $n = p_1^{a_1} \cdots p_r^{a_r}$, 其中

$r \in \mathbb{Z}_{\geq 0}$ ($r=0$ 对应于 $n=1$ 的情形), $p_1, \dots, p_r$ 是相异素数, $a_1, \dots, a_r \in \mathbb{Z}_{\geq 1}$.

$$\text{则有: } \varphi(n) = n \prod_{i=1}^r \left(1 - \frac{1}{p_i}\right) = \prod_{i=1}^r (p_i^{a_i} - p_i^{a_i-1})$$

Proof: 当 $n=1$ 时, 此时 $r=0$

$$\therefore \text{左边} = \varphi(1) = 1, \quad \text{中间} = 1 \times \prod_{i=1}^0 \left(1 - \frac{1}{p_i}\right) = 1 \times 1 = 1. \quad \text{右边} = \prod_{i=1}^0 (p_i^{a_i} - p_i^{a_i-1}) = 1$$

$$\therefore \text{左边} = \text{中间} = \text{右边}.$$

下设 $n \in \mathbb{Z}_{\geq 2}$. 此时 $r \in \mathbb{Z}_{\geq 1}$.

$$\text{右边} = \prod_{i=1}^r (p_i^{a_i} - p_i^{a_i-1}) = \prod_{i=1}^r p_i^{a_i} \left(1 - \frac{p_i^{a_i-1}}{p_i^{a_i}}\right) = \prod_{i=1}^r p_i^{a_i} \left(1 - \frac{1}{p_i}\right)$$

$$= \left(\prod_{i=1}^r p_i^{a_i}\right) \cdot \left(\prod_{i=1}^r \left(1 - \frac{1}{p_i}\right)\right) = (p_1^{a_1} \cdots p_r^{a_r}) \cdot \prod_{i=1}^r \left(1 - \frac{1}{p_i}\right) = n \prod_{i=1}^r \left(1 - \frac{1}{p_i}\right) = \text{中间}.$$

记 $A_i = \{d \in \mathbb{Z}_{\geq 1} : d \leq n \text{ 且 } p_i | d\}$, $i=1, \dots, r$.

对 $\forall d \in \mathbb{Z}_{\geq 1}$ 且 $d \leq n$, 有:

若 $\gcd(d, n) \neq 1$, 则 $p_1 | d$ 或 $p_2 | d$ 或 $\dots$ 或 $p_r | d$
 $\therefore d \in A_1$ 或 $d \in A_2$ 或 $\dots$ 或 $d \in A_r \quad \therefore d \in \bigcup_{i=1}^r A_i$

若 $d \in \bigcup_{i=1}^r A_i$, 则 $d \in A_1$ 或 $d \in A_2$ 或 $\dots$ 或 $d \in A_r$

$\therefore p_1 | d$ 或 $p_2 | d$ 或 $\dots$ 或 $p_r | d \quad \therefore \gcd(d, n) \neq 1$

$\therefore$ 对 $\forall d \in \mathbb{Z}_{\geq 1}$ 且 $d \leq n$, 有: $\gcd(d, n) \neq 1 \Leftrightarrow d \in \bigcup_{i=1}^r A_i$

$\therefore$ 对 $\forall d \in \mathbb{Z}_{\geq 1}$ 且 $d \leq n$, 有: $\gcd(d, n) = 1 \Leftrightarrow d \notin \bigcup_{i=1}^r A_i$

~~$\therefore \varphi(n) =$~~ $\therefore$ 对 $\forall i=1, \dots, r$, 有: $A_i = \{d \in \mathbb{Z}_{\geq 1} : d \leq n \text{ 且 } p_i | d\} \subseteq \{1, \dots, n\}$

$\therefore \bigcup_{i=1}^r A_i \subseteq \{1, \dots, n\}$

$\therefore$ 对 $\forall d \in \mathbb{Z}_{\geq 1}$ 且 $d \leq n$, 有: $\gcd(d, n) = 1 \Leftrightarrow d \in \{1, \dots, n\} \setminus \bigcup_{i=1}^r A_i$

$\therefore \varphi(n) = |\{1, \dots, n\} \setminus \bigcup_{i=1}^r A_i| = |\{1, \dots, n\}| - |\bigcup_{i=1}^r A_i|$

$= n - \sum_{k=1}^r (-1)^{k+1} \left(\sum_{1 \leq i_1 < \dots < i_k \leq r} |A_{i_1} \cap \dots \cap A_{i_k}| \right)$

~~$= n - \sum_{k=1}^r (-1)^{k+1} \dots$~~

$= n - \sum_{k=1}^r (-1)^{k+1} \left(\sum_{1 \leq i_1 < \dots < i_k \leq r} \frac{n}{p_{i_1} \dots p_{i_k}} \right)$

$$= n - \sum_{k=1}^r (-1)^{k+1} \cdot n \cdot \left(\sum_{1 \leq i_1 < \dots < i_k \leq r} \frac{1}{p_{i_1} \dots p_{i_k}} \right)$$

$$= n \left(1 - \sum_{k=1}^r (-1)^{k+1} \left(\sum_{1 \leq i_1 < \dots < i_k \leq r} \frac{1}{p_{i_1} \dots p_{i_k}} \right) \right)$$

$$= n \left(1 + \sum_{k=1}^r (-1)^k \left(\sum_{1 \leq i_1 < \dots < i_k \leq r} \frac{1}{p_{i_1} \dots p_{i_k}} \right) \right)$$

$$= n \left(1 - \frac{1}{p_1} \right) \left(1 - \frac{1}{p_2} \right) \dots \left(1 - \frac{1}{p_r} \right) = n \prod_{i=1}^r \left(1 - \frac{1}{p_i} \right)$$

∴ 左边 = 中间. $\square$

Remark:

① 对于 $k \in \{1, \dots, r\}$, $1 \leq i_1 < \dots < i_k \leq r$, 为什么 $|A_{i_1} \cap \dots \cap A_{i_k}| = \frac{n}{p_{i_1} \dots p_{i_k}}$?

≡ 对 $\forall d \in \mathbb{Z}_{\geq 1}$ 且 $d \leq n$:

若 $p_{i_1} \dots p_{i_k} \mid d$, 则有: $p_{i_1} \mid d$ 且 $\dots$ 且 $p_{i_k} \mid d$, $\therefore d \in A_{i_1}$ 且 $\dots$ 且 $d \in A_{i_k}$

$\therefore d \in A_{i_1} \cap \dots \cap A_{i_k}$.

若 $d \in A_{i_1} \cap \dots \cap A_{i_k}$, 则 $d \in A_{i_1}$ 且 $\dots$ 且 $d \in A_{i_k}$

$\therefore p_{i_1} \mid d$ 且 $\dots$ 且 $p_{i_k} \mid d \quad \because p_{i_1}, \dots, p_{i_k}$ 全不为 0, $p_{i_1}, \dots, p_{i_k}$ 两两互素

$\therefore p_{i_1} \dots p_{i_k} \mid d$

$\therefore$ 对 $\forall d \in \mathbb{Z}_{\geq 1}$ 且 $d \leq n$, 有: $d \in A_{i_1} \cap \dots \cap A_{i_k} \iff p_{i_1} \dots p_{i_k} \mid d$

$\therefore A_{i_1} \cap \dots \cap A_{i_k}$ 中的元素是: $p_{i_1} \dots p_{i_k} \times 1, p_{i_1} \dots p_{i_k} \times 2, \dots, p_{i_1} \dots p_{i_k} \times \frac{n}{p_{i_1} \dots p_{i_k}}$

$\therefore |A_{i_1} \cap \dots \cap A_{i_k}| = \frac{n}{p_{i_1} \dots p_{i_k}}$

② $(1 - \frac{1}{p_1})(1 - \frac{1}{p_2}) \cdots (1 - \frac{1}{p_r})$ 展开后长什么样?

$$(1 - \frac{1}{p_1})(1 - \frac{1}{p_2}) \cdots (1 - \frac{1}{p_r})$$

$$= 1 + (-1)^1 \sum_{i=1}^r \frac{1}{p_i} + (-1)^2 \sum_{1 \leq i < j \leq r} \frac{1}{p_i p_j} + (-1)^3 \sum_{1 \leq i < j < k \leq r} \frac{1}{p_i p_j p_k} \\ + \cdots + (-1)^r \sum_{1 \leq i < \cdots < r \leq r} \frac{1}{p_i \cdots p_r}$$

$$= 1 + \sum_{k=1}^r (-1)^k \left(\sum_{1 \leq i_1 < \cdots < i_k \leq r} \frac{1}{p_{i_1} \cdots p_{i_k}} \right)$$